

Homework 1

Due: Friday, February 11

All homeworks are due at 11:55 PM on Gradescope.

Please do not include any identifying information about yourself in the handin, including your Banner ID.

Be sure to fully explain your reasoning and show all work for full credit.

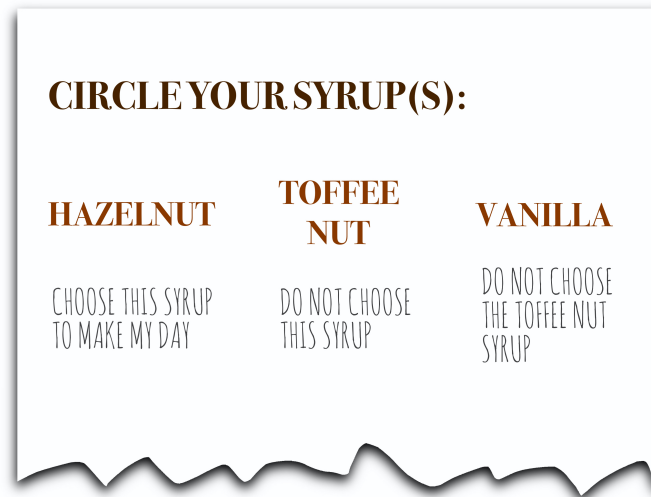
Problem 1

At a local coffee shop, there are five coffee toppings: cold foam, caramel drizzle, whipped cream, cinnamon dolce sprinkles, and cocoa powder.

If cold foam isn't chosen, caramel drizzle is chosen. If whipped cream and caramel drizzle both are chosen, then cocoa powder can't be chosen. If cinnamon dolce sprinkles isn't chosen then cocoa powder must be chosen.

- a. Use one variable to represent choosing each coffee topping, and write logical expressions using $\wedge$, $\vee$, $\rightarrow$, and $\neg$ to demonstrate the above paragraph.
- b. If neither cinnamon dolce sprinkles nor cold foam is chosen, what is? What isn't? Explain your logic.
- c. It's Pete's first day working at this local coffee shop! His first customer arrives and passes Pete a slip of paper with his coffee order. After venturing to the back of the shop, he discovers three different types of syrups: hazelnut, toffee nut, and vanilla.

One of the syrups will make the customer extremely happy if added, while the other two syrups will give his customer an allergic reaction. On the customer's order, there is a message below each of the syrup options:



Given that at least one of the messages is true, and at least one of them is false, which syrup should be used to ensure that Pete's customer *doesn't* get an allergic reaction?

Solution:

a.

- f: Cold foam is chosen
- d: Caramel drizzle is chosen
- w: Whipped cream is chosen
- s: Cinnamon dolce sprinkles is chosen
- p: Cocoa powder is chosen

$$\neg f \rightarrow d \quad (1)$$

$$(w \wedge d) \rightarrow \neg p \quad (2)$$

$$\neg s \rightarrow p \quad (3)$$

b.

Cocoa powder is chosen (justified by expression 3 above), and caramel drizzle is chosen (justified by expression 1 above). Whipped cream cannot possibly have been chosen because cocoa powder is, and expression 2 above would contradict that.

c.

Let us use the following propositional statements:

- h : the hazelnut syrup should be added
- t : the toffee nut syrup should be added
- v : the vanilla syrup should be added

From the problem description, we know that exactly one syrup will make the customer happy, and the other two will give the customer an allergic reaction. This allows us to obtain the following:

- $(h \wedge \neg t \wedge \neg v) \vee (\neg h \wedge t \wedge \neg v) \vee (\neg h \wedge \neg t \wedge v)$

Consequently, we only need to observe scenarios in which exactly one of the listed propositional statements is true, with the other two being false.

Additionally, from the problem description, we know that at least one of the messages below the three syrup options must be true. Based on the messages, we can, subsequently, obtain the following:

- $A: (h \vee \neg t)$

Finally, from the problem description, we also know that at least one of the messages on the three buttons must be false. Based on the messages, we can, subsequently, obtain the following:

- $B: (\neg h \vee t)$

From A and B , and knowing we only need to observe scenarios in which exactly one of the listed propositional statements is true, we can create the following truth table:

h	t	v	A	B	$A \wedge B$
T	F	F	T	F	F
F	T	F	F	T	F
F	F	T	T	T	T

Consequently, the vanilla syrup should be used!

Rubric:

1.5 points for defining variables (a.); deduct 0.5 if they just say "Cold foam" rather than "Cold foam is chosen", for example.

0.5 points for each correct logical expression (1.5 total). (a.)

0.5 points for each correct deduction (1.5 total). (b.)

0.5 points for each explanation (1.5 total). (b.)

1 point for correct answer. (c.)

3 points for correct reasoning. (c.)

Total points: $3 + 3 + 4 = 10$

Problem 2

For each of the following expressions, convert it to both CNF and DNF using the provided method (show all work!).

a. Expression: $(p \rightarrow q) \rightarrow r$

Method: Rewrite rules (hint: do DNF first)

b. Expression: $p \rightarrow (q \rightarrow r)$

Method: Truth tables

Solution:

- a. Short CNF:
- $(r \vee p) \wedge (r \vee \neg q)$

Short DNF: $(p \wedge \neg q) \vee (r)$

p	q	r	$(p \rightarrow q) \rightarrow r$	CNF	DNF
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	F	F	F
F	F	T	T	T	T
F	F	F	F	F	F

Rewrite for DNF

$$(p \rightarrow q) \rightarrow r$$

iff $\neg(\neg p \vee q) \vee r$ (by Definition of Conditional twice)iff $(\neg\neg p \wedge \neg q) \vee r$ (by De Morgan's Law)iff $(p \wedge \neg q) \vee r$ (by Double Negative Law)

Rewrite for CNF

$$(p \rightarrow q) \rightarrow r$$

iff $(p \wedge \neg q) \vee r$ (shown above)iff $r \vee (p \wedge \neg q)$ (by Commutative Law)iff $(r \vee p) \wedge (r \vee \neg q)$ (by Distributive Law)

- a. CNF:
- $(\neg p \vee \neg q \vee r)$

Expanded DNF: $(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$

p	q	r	$p \rightarrow (q \rightarrow r)$	CNF	DNF
T	T	T	T	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	T	T	T

Rubric:

- a. 1 point for each correct formula
2 points for each correct proof
(6 points total)
- b. 1 point for each correct formula
2 points for correct truth table
(4 points total)

(10 points total)

Problem 3

- a. Although we have been using AND and OR gates with only two inputs, we can create AND and OR gates that take in more than two inputs. Under the hood, these can just be implemented with 2-input AND and OR gates. Build a 3-input AND circuit using only 2-input AND gates, and build a 3-input OR circuit using only 2-input OR gates.

- b. Duncan and Bucky have spent many nights arguing about who makes better lattes. They decide to create a poll to determine what people think. However, to do so, they need a machine that can handle the poll data.

To help them out, construct a circuit using only AND, OR, and NOT gates. You may use gates of any number of inputs for this problem.

The circuit will take in three people's answers (1 for Duncan and 0 for Bucky), and return the answer of the majority.

Please additionally explain how your circuit works.

- c. Joe and Pete are currently having a lot of trouble determining if the numbers 0, 1, 2, and 3 are odd or even. The solution? Build a circuit!

Model this problem as a circuit using only AND, OR, and NOT gates. You may use gates with more than one input. Create three input wires and one output wire. If an even number of inputs are on, the output should be off. If an odd number of input wires are on, the output should be on.

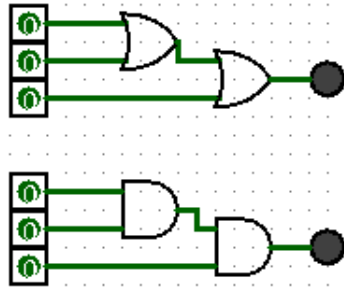
Please additionally explain how your circuit works.

Note: The output depends only on how many of the input wires are on. For example, if exactly one input is on, then the output should be on. It does not matter which input is on.

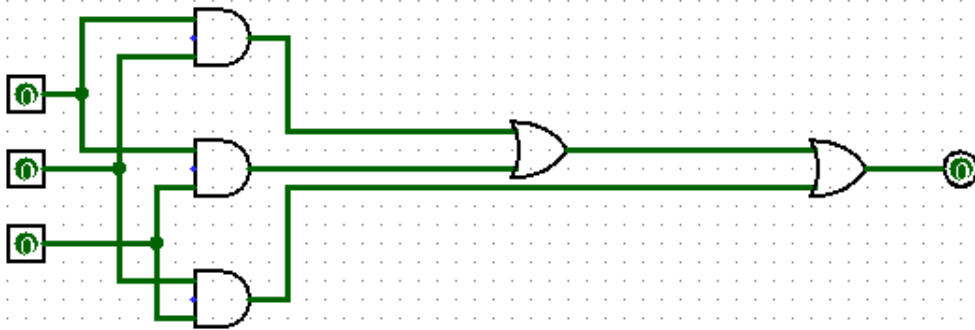
- x. Food for thought (not for points): What happens if you combined your circuit from b. with your circuit from c.? Think about the two individual outputs as a single, two piece output.

Solution:

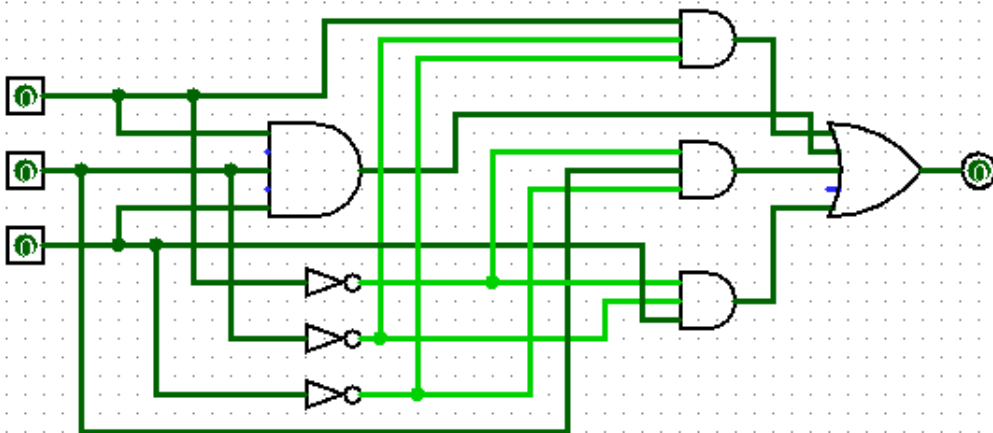
Part (a)



Part (b)



Part (c)



Explanation for part b: We should return 1 if and only if a majority of inputs are 1, which is true if and only if at least two inputs are 1, which is true if and only if

any two inputs are 1. So, it is sufficient to create a circuit that checks if any two inputs are 1. We thus generate an expression where every pair of inputs is checked: $(p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$. If any two inputs are 1, then one of these will turn up 1, and so the result will be 1.

Explanation for part c: We should return 1 if and only if the number of inputs that are 1 is odd, which is true if and only if all three inputs are 1 or exactly one input is 1. We thus generate an expression where each of these scenarios is checked: $(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r)$ checks if there is exactly one wire that is 1 (one clause for each wire), and $(p \wedge q \wedge r)$ checks if all three wires are 1. So, we construct a circuit for the expression $(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge q \wedge r)$.

Rubric:

1 point for correctness (a.)

1.5 points for correctness (b.)

1.5 point for explanation (b.)

2 points for correctness (c.)

2 point for explanation (c.)

Problem 4

Challenge Problem! (Extra credit)

The n -queens problem involves a $n \times n$ chess board and n queens. The goal is to place all of the queens on the board so that no queen is “threatening” another queen (able to attack it). Queens can move as far as they want vertically, horizontally, or diagonally.

a. For $n = 1$ to 4, either construct a solution to the n -queens problem, or (briefly) justify why one does not exist.

b. For an arbitrary positive integer n , describe how to construct a boolean formula which is satisfiable if and only if the n -queen problem is solvable. Your description should be explicit enough that anyone who has read the problem can follow them.

Hint: Think about what your variables in the formula will represent (What does it mean if a given variable is true? If it’s false?).

Hint 2: Break the problem into two pieces. How can we enforce that no queens threaten each other? Given that no queens threaten each other, how can we enforce

that there are n queens?

Solution:

- a. There are solutions for $n = 1$ and $n = 4$, but not for $n = 2$ or $n = 3$.
- b. For a given n , we use variables $x_{(1,1)}$ through $x_{(n,n)}$, where variable $x_{(i,j)}$ represents whether a queen is present on spot (i, j) . We construct a formula f_n which will be true if and only if there is a solution to the n -queens problem. Let $f_n = (h_n \wedge v_n \wedge d_n) \wedge k_n$, where:
- a. h_n enforces that no two queens attack each other horizontally by conjoining over clauses $\neg(x_{(i_1,j)} \wedge x_{(i_2,j)})$ (with $i_1 \neq i_2$)
 - b. h_n enforces that no two queens attack each other vertically by conjoining over clauses $\neg(x_{(i,j_1)} \wedge x_{(i,j_2)})$ (with $j_1 \neq j_2$)
 - c. d_n enforces that no two queens attack each other diagonally by conjoining over clauses $\neg(x_{(i_1,j_1)} \wedge x_{(i_2,j_2)})$ (with $(i_1, j_1) \neq (i_2, j_2)$ and $|i_1 - i_2| = |j_1 - j_2|$)
 - d. k_n enforces that all n queens are present by checking for a queen in every column:

$$\bigwedge_{i \in [1,n]} \bigvee_{j \in [1,n]} x_{(i,j)} = ((x_{(1,1)} \vee x_{(1,2)} \vee \dots) \wedge (x_{(2,1)} \vee x_{(2,2)} \vee \dots) \wedge \dots)$$

Rubric:

- 0.5 point each for solutions to $n = 1$ through 4. (a.)
- 1 point for coming up with good variable representation. (b.)
- 3 points for enforcing that no queens threaten each other (with explanation). (b.)
- 2 points for enforcing that all n queens are present (with explanation). (b.)